

Project Domain

Domain: Artificial Intelligence (AI) / Machine Learning (ML) in Banking and Financial Technology (FinTech)

The proposed project belongs to the domain of Artificial Intelligence and Machine Learning with applications in Banking and Financial Technology (FinTech). It focuses on credit risk assessment, a critical process used by banks and financial institutions to evaluate whether a loan applicant is likely to repay a loan or become a defaulter. Traditional credit evaluation methods rely on manual verification and rule-based scoring, which may be time-consuming, inconsistent, and less effective when dealing with large and complex financial datasets. By applying machine learning algorithms, the system can analyze customer financial information such as income, employment status, credit history, repayment behavior, existing liabilities, and loan details to predict the probability of loan default with higher accuracy. The project also incorporates Explainable Artificial Intelligence (XAI) using SHAP to provide transparent explanations for every prediction, helping both customers and financial institutions understand the factors influencing the risk score. In addition to predicting credit risk, the system recommends a safe loan amount and affordable EMI based on the applicant's financial capacity, making it a practical decision-support tool for banks. This project demonstrates how AI can improve financial decision-making, reduce non-performing loans (NPLs), enhance transparency, and support faster, fairer, and more reliable loan approval processes. It represents a real-world application of machine learning in the banking sector and contributes to the growing adoption of intelligent financial services.